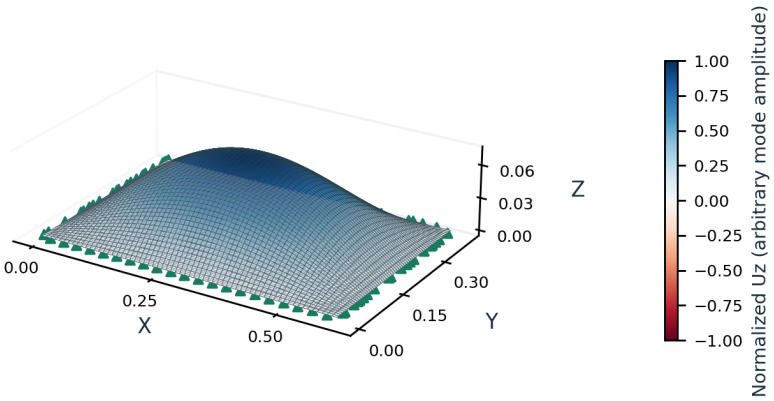
Symmetric cross-ply plate: Navier vibration

ACA04 | SOL 103 | Independent analytical equation verification

The same symmetric cross-ply plate tests laminate stiffness and mass through six natural frequencies. Independent classical laminate integration supplies the Navier spectrum and sinusoidal mode families.

Geometry / mesh	Plate 0.6 m by 0.4 m, thickness 2 mm; CQUAD4 meshes from 4x4 to 64x64.
Material / section	[0/90/90/0], four 0.5 mm plies; E1 = 135 GPa, E2 = 10 GPa, G12 = 5 GPa, nu12 = 0.3; density 1600 kg/m3.
Boundary conditions	All edges uz=0; bending rotations free. ux=0 on x=0; origin uy=0; drilling rotation constrained.
Applied load	Free vibration; no applied load. Material density included in each ply.

Mode 1 | 35.018 Hz



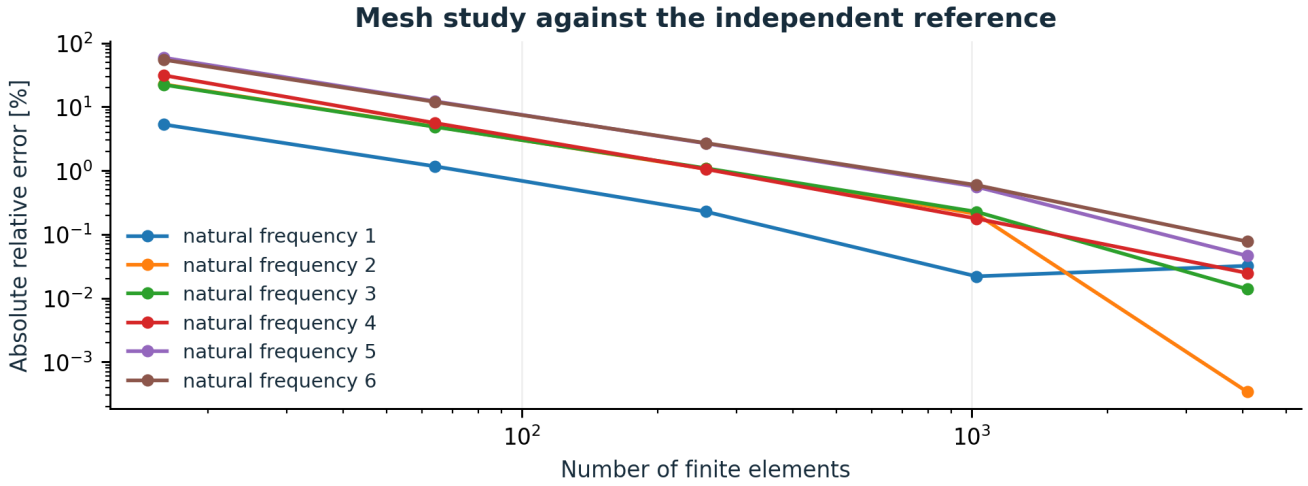
Shape amplified x0.072; grey = original. Eigenvector sign and amplitude are arbitrary.

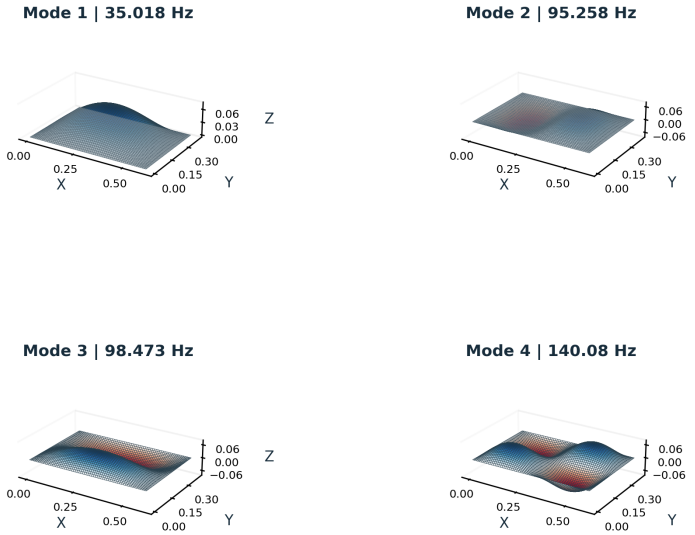
Actual solver output. Deformation is amplified for visibility; mode vectors have arbitrary sign and amplitude.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
natural frequency 1 [Hz]	35.018	35.029	0.032114%	PASS
natural frequency 2 [Hz]	95.258	95.258	0.00033892%	PASS
natural frequency 3 [Hz]	98.473	98.459	0.013829%	PASS
natural frequency 4 [Hz]	140.08	140.12	0.024731%	PASS
natural frequency 5 [Hz]	203.12	203.03	0.046038%	PASS
natural frequency 6 [Hz]	211.17	211.01	0.077249%	PASS

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

Independent reference equations

D_ij=sum_k Qbar_ij(k)*(z_k^3-z_(k-1)^3)/3

alpha=m*pi/a; beta=n*pi/b

H=D11*alpha^4+2*(D12+2D66)*alpha^2*beta^2+D22*beta^4

f_mn=sqrt(H/(rho*t))/(2*pi)

All first six finest frequencies agree within 0.07725%; minimum transverse mode-shape MAC = 0.999999947.

The worst error falls from 0.5918% at 32x32 to 0.07725% at 64x64; coarse results are retained.

Scope: Both reference and FE model omit independent rotary inertia. Thin-theory differences arise from transverse-shear flexibility and mesh discretization.

Reference and reproduction

Leissa, Vibration of Plates, NASA SP-160 (1969) https://ntrs.nasa.gov/citations/19700009156

Nachiketa Tiwari / IIT Kanpur: Mechanics of Laminated Composite Structures, Lecture37 Buckling of Plates
https://archive.nptel.ac.in/content/storage2/courses/112104161/L37.pdf

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997cce40 | Full precision values: results.json / metrics.csv